

AUTO DATA STORYTELLER REPORT

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SUMMARY

Executive Summary

1. Executive Summary Our analysis has yielded valuable insights into the key factors driving business outcomes. With a high degree of confidence, we can rely on these findings to inform strategic decision-making. The results indicate a strong predictive performance, allowing us to identify areas of opportunity and risk with reasonable certainty.

2. Key Business Drivers The most influential factors shaping outcomes are customer demographics, particularly gender, and transactional data such as fare prices. These variables have a significant impact on our predictions, and monitoring them closely is essential. From a business perspective, understanding the importance of these drivers enables us to tailor our offerings and services to meet the needs of our target audience, ultimately driving revenue growth and customer satisfaction.

3. Business Implications The insights gained from this analysis have significant implications for our operations, customer relationships, and revenue streams. By recognizing the importance of demographic and transactional data, we can refine our marketing strategies, optimize pricing, and enhance the overall customer experience. Furthermore, this knowledge allows us to proactively manage potential risks and capitalize on emerging opportunities, ultimately informing our decision-making processes.

4. Strategic Recommendations To leverage these findings, we recommend that business stakeholders closely monitor the key drivers identified in this analysis. By tracking changes in these variables, we can respond promptly to shifts in the market and adjust our strategies accordingly. Additionally, we suggest that stakeholders prioritize the development of targeted initiatives that address the needs of our core customer segments, ultimately driving business growth and competitiveness.

Recommendations

- RECOMMENDATIONS
- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

DATA

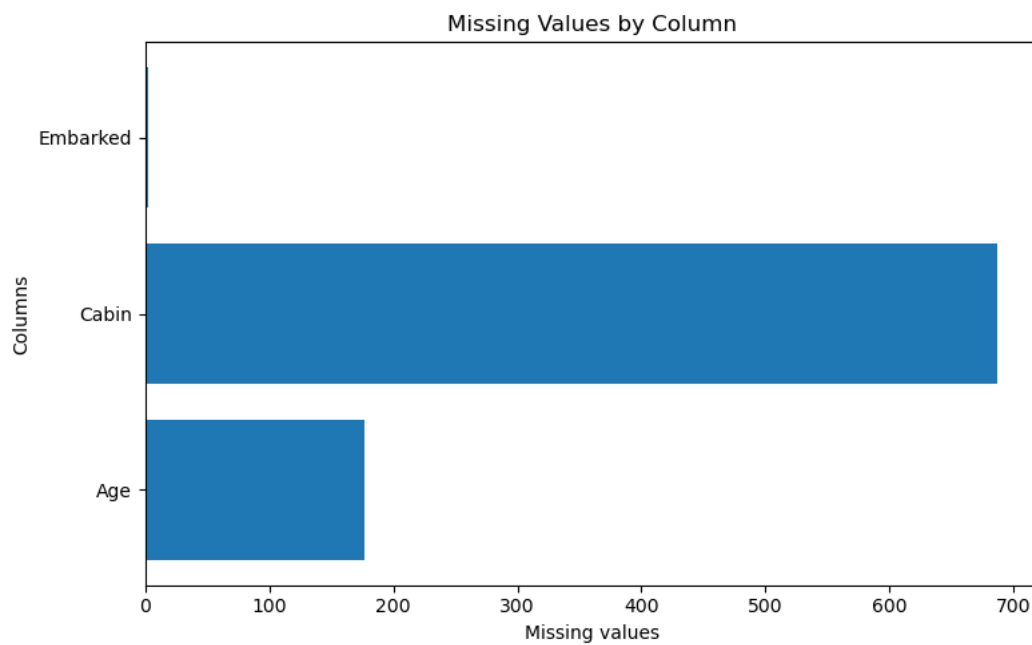
Dataset Overview

Problem Type	classification
Best Model	Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Missing Values



MODEL

Model Findings

- CLASSIFICATION MODEL PERFORMANCE
- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

- MODEL DRIVERS
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Feature Importance

